

INNER JOIN Example

Goal: Show film titles and their languages.

```
SELECT film.title, language.name
```

```
FROM film
```

```
INNER JOIN language ON film.language_id = language.language_id;
```

Explanation:

- INNER JOIN returns only rows that have matching language_id in both film and language.
- Every film has a language, so all rows will appear.

Sample Output:

title	name
ACADEMY DINOSAUR	English
ACE GOLDFINGER	English
...	...

LEFT JOIN Example

Goal: Show all customers and their rental records (if any).

```
SELECT customer.customer_id, customer.first_name, rental.rental_id
```

```
FROM customer
```

```
LEFT JOIN rental ON customer.customer_id = rental.customer_id;
```

Explanation:

- Returns **all customers**, even those who **never rented** a movie.
- If no rental found, rental_id will be NULL.

Sample Output:

customer_id	first_name	rental_id
1	MARY	12345
2	PATRICIA	NULL

RIGHT JOIN Example

Goal: Show all rentals and their customers (if any).

```
SELECT customer.customer_id, customer.first_name, rental.rental_id
FROM customer
RIGHT JOIN rental ON customer.customer_id = rental.customer_id;
```

Explanation:

- Returns **all rentals**, even if a rental doesn't have a valid customer (rare case).
- RIGHT JOIN is the opposite of LEFT JOIN.

Sample Output:

customer_id	first_name	rental_id
1	MARY	12345
NULL	NULL	67890

FULL JOIN Example (MySQL workaround)

MySQL doesn't support FULL JOIN directly, but you can simulate it using UNION.

Goal: Combine results of LEFT and RIGHT JOIN to show all customers and rentals (matched or unmatched).

```
SELECT customer.customer_id, customer.first_name, rental.rental_id
FROM customer
LEFT JOIN rental ON customer.customer_id = rental.customer_id
UNION
SELECT customer.customer_id, customer.first_name, rental.rental_id
FROM customer
RIGHT JOIN rental ON customer.customer_id = rental.customer_id;
```

Explanation:

- LEFT JOIN shows all customers.
- RIGHT JOIN shows all rentals.
- UNION merges both → this mimics a **FULL OUTER JOIN**.

Sample Output:

customer_id	first_name	rental_id
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1	MARY	12345
2	PATRICIA	NULL
NULL	NULL	67890

Practice Set

No.	Task	Hint
1	Show films and their categories	Use film + film_category
2	List all stores and the staff working there	Use store + staff
3	Show all customers and the city they live in	Use customer + address
4	List all payments and the customers who made them	Use payment + customer
5	Show all films and their rental records (some films may not be rented)	Use film + inventory with LEFT JOIN